

Exp No: 26

TRANSACTION LEVEL MODELING (TLM) USING SYSTEMC

```
GNU nano 8.7.1 tlm.cpp *
#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>

using namespace sc_core;
using namespace tlm;
using namespace std;

// Target
struct Target : sc_module {
    tlm_utils::simple_target_socket<Target> socket;

    SC_CTOR(Target) : socket("socket") {
        socket.register_b_transport(this, &Target::b_transport);
    }

    void b_transport(tlm_generic_payload& trans, sc_time& delay) {
        int* data = (int*) trans.get_data_ptr();

        cout << "Target received: " << *data << endl;

        delay += sc_time(10, SC_NS);
    }
};

// Initiator
struct Initiator : sc_module {
    tlm_utils::simple_initiator_socket<Initiator> socket;
};
```

Write to File: tlm.cpp

^G Help	M-D DOS Format	M-A Append	M-B Backup File	^T Browse
^C Cancel	M-M Mac Format	M-P Prepend	^Q Discard buffer	

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```
GNU nano 8.7.1 CMakeLists.txt
cmake_minimum_required(VERSION 3.10)

project(SystemC_TLM)

set(CMAKE_CXX_STANDARD 17)

include_directories(/usr/include)

add_executable(tlm tlm.cpp)

target_link_libraries(tlm systemc)
```

Write to File: CMakeLists.txt

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^C Cancel	M-M Mac Format	M-P Prepend	^Q Discard buffer	

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OUTPUT:

```
sai_varshini@DESKTOP-5G7BJMJ x + v
add_executable(tlm tlm.cpp)

target_link_libraries(tlm SystemC::systemc)
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test$ nano CMakeLists.txt
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test$ cat CMakeLists.txt
cmake_minimum_required(VERSION 3.10)

project(SystemC_TLM)

set(CMAKE_CXX_STANDARD 17)

include_directories(/usr/include)

add_executable(tlm tlm.cpp)

target_link_libraries(tlm systemc)
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test$ cd build
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/build$ cmake ..
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/sai_varshini/systemc_test/build
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/build$ make
[ 50%] Building CXX object CMakeFiles/tlm.dir/tlm.cpp.o
[100%] Linking CXX executable tlm
[100%] Built target tlm
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/build$ ./tlm

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Initiator sending: 555
Target received: 555
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/build$ |
```